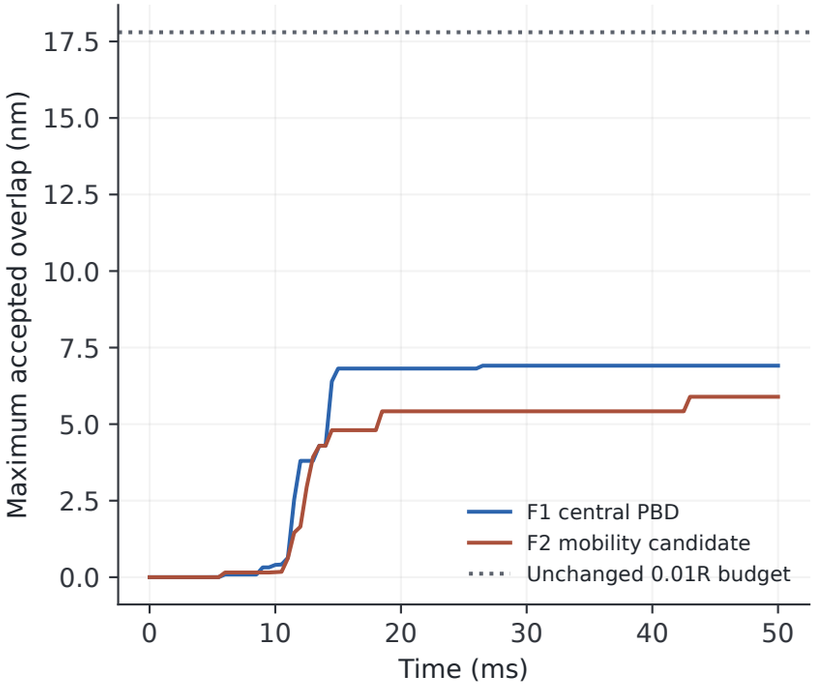
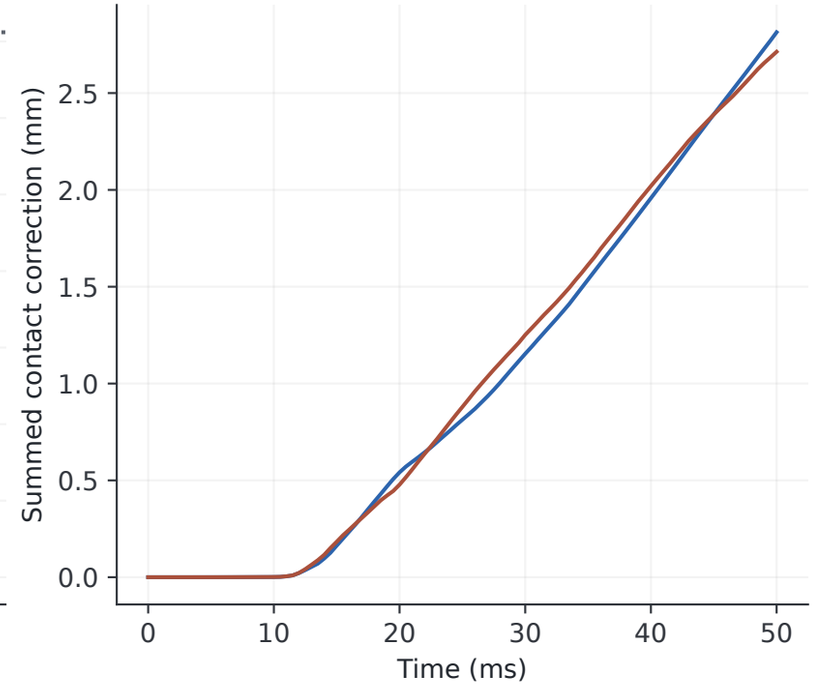
F2-2 | Contact and packing sensitivity
True radius and identical 3D view in d–e · force model fixed · EXPLORATORY

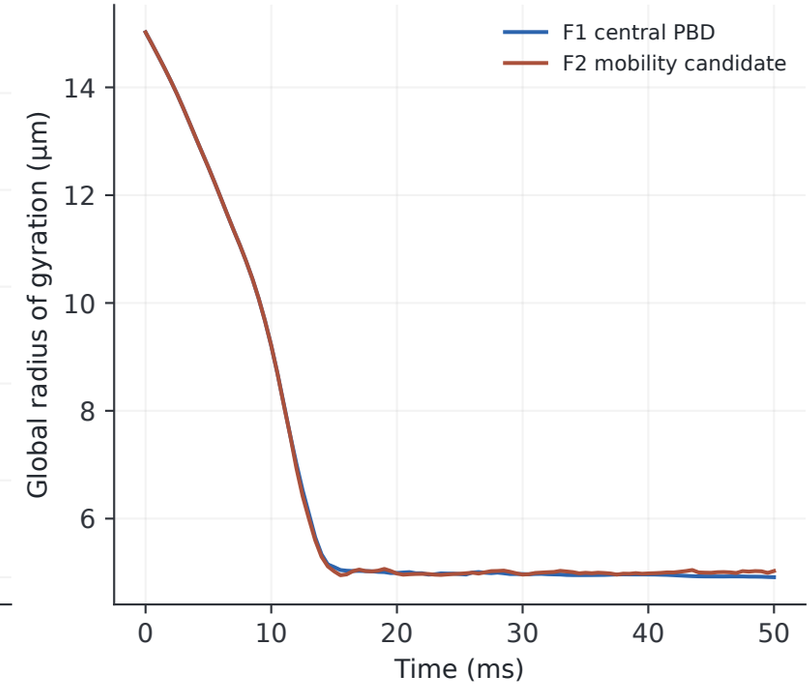
a Cumulative maximum, matched 5 µs



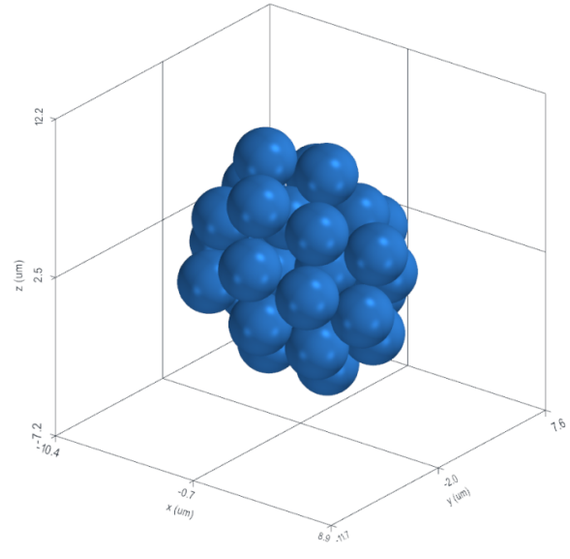
b Accumulated over all particles



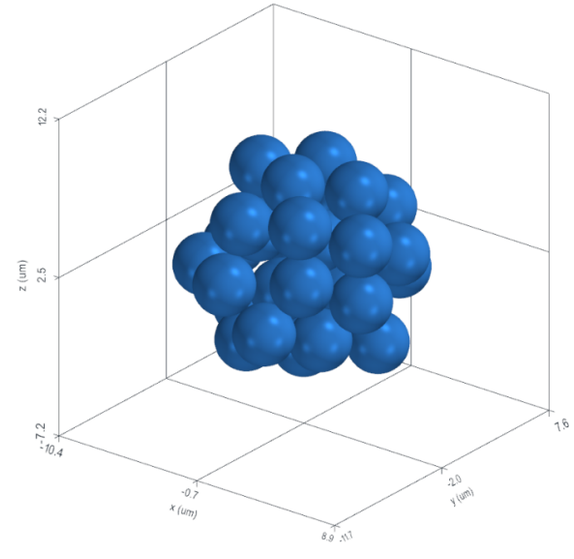
c Same force; distinct packing



d F1 central PBD · 50 ms



e F2 mobility candidate · 50 ms



f Matched comparison and disposition

Time to Nmax ≥ 16: 10.5 / 10.5 ms
Final components: 1 / 1 (32 bubbles)
Final Rg difference: 2.34%
Eigenvalue difference: 12.45%
ID-aligned RMSD: 3.67 µm

CONTACT_MODEL_SENSITIVE
Candidate rejected as default:
noncentral corrections induce rotation.

Secondary OFF: no motion or aggregation.
Final packing remains unvalidated.